

Problem Description

You are given an N -dimensional array A . You can execute an operation that removes the greatest and smallest elements from the array and replaces them with their difference. As a result, the array's size will drop by one after each operation. You are assigned Q jobs, each of which contains an integer K . After K operations, you must provide the total of all the items in the array for each task.

Constraints:

$$2 \leq N \leq 10^5$$

$$1 \leq Q \leq 10^5$$

$$0 \leq A[i] \leq 10^9$$

$$0 \leq K < N$$

Input:

First line contains two space-separated integers N and Q , denoting the number of elements in array and number of queries respectively.

Next line contains N space-separated integers denoting elements of the array.

Next Q lines contain a single integer K .

Output:

For each task, print answer in a new line.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

5 2
3 2 1 5 4
1
2

EXPECTED OUTPUT

13
9

Test Case 2

INPUT (STDIN)

5 3
13 12 11 15 14
1
2
3

EXPECTED OUTPUT

43
35
15

▼ Mandatory Test Cases

Test Case 1

KEYWORD

for(i=0;i<n;i++)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

4

Test Case 2

TOKEN COUNT

300

Test Case 3

NLOC

38